

Integrals – 3

Differentiation under the integral sign and Frullani integrals

Homework

1 сентября 2026 года

Theoretical minimum. The Leibniz rule and its hypotheses; uniform convergence of an improper integral in a parameter; Frullani's theorem; the regularising factor $e^{-\alpha x}$; interchanging the order of integration (Fubini) as a second route to the same answers.

1. Compute $\int_0^\infty e^{-\alpha x} \frac{\sin(\beta x)}{x} dx$ for $\alpha > 0$ and deduce the value of $\int_0^\infty \frac{\sin(\beta x)}{x} dx$ depending on the sign of β .
2. Compute $\int_0^1 \frac{\arctan(\alpha x)}{x\sqrt{1-x^2}} dx$ by differentiating in α .
3. Compute $\int_0^\infty \frac{\ln(1+\alpha^2 x^2)}{1+x^2} dx$ and check the answer at $\alpha = 1$ by an independent argument.
4. Construct an example in which formal differentiation under the integral sign gives a wrong answer, and point out exactly which hypothesis of the theorem fails.
5. Compute $\int_0^\infty \frac{e^{-x} - e^{-x \cosh t}}{x} dx$ as a function of t , and explain why the answer does not depend on the way the integrand is split.
6. By interchanging the order of integration compute $\int_0^1 \frac{x-1}{\ln x} dx$ and $\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} \cos(cx) dx$.
7. Compute $\int_0^{\pi/2} \ln(a^2 \cos^2 x + b^2 \sin^2 x) dx$ for $a, b > 0$ by differentiating in a parameter.
8. (Integration Bee) Integrals with a parameter hidden in the exponent:
 $\int_0^2 2^{\log x} dx$ (2019), $\int_0^e x^{1/\log x} dx$ (2019), $\int (2015)^x dx$ (2015) and $\int e^{e^{2016x} + 6048x} dx$ (2016).

9. Compute $\int_0^\infty \frac{\arctan(ax) - \arctan(bx)}{x} dx$ and explain why this is a special case of Frullani's theorem even though $\arctan$ does not tend to zero at infinity.
10. Compute $\int_0^\infty \frac{\sin^2 x}{x^2} dx$ in three ways: by parts, reducing it to $\int \frac{\sin x}{x}$; by differentiating $\int_0^\infty \frac{\sin^2(\alpha x)}{x^2} dx$ in the parameter; and through Parseval's identity.